Linear Regression: Concepts & Worked Example

Lecture 2 Knowledge Check Solutions | Instructor: Assoc. Prof. Mohamed Abd Elfattah (MET)

Question 1 of 8

In $\hat{y} = \beta_0 + \beta_1 x$, what does β_1 represent?

A) The predicted value when x = 0

B) The change in $\hat{y}$ for a one-unit increase in x

C) The average of y

D) The error of the model

Answer: B — The slope β_1 measures how much the predicted output $\hat{y}$ changes when the input feature x increases by one unit.

Question 2 of 8

Linear regression is used to predict:

A) A discrete category

B) A cluster label

C) A continuous numeric value

D) A reward

Answer: C — Linear regression is a supervised learning algorithm designed for regression tasks to forecast continuous numeric outcomes.

Question 3 of 8

A residual is defined as:

A) $\hat{y} - y$

B) $y - \hat{y}$

C) $x - \bar{x}$

D) $\beta_1 - \beta_0$

Answer: B — The residual (e) is the difference between the actual observed value (y) and the predicted value from the model ($\hat{y}$).

Question 4 of 8

Linear regression chooses the line that minimizes:

A) The sum of squared residuals

B) The number of data points

C) The slope

D) The largest x value

Answer: A — The least-squares method finds the optimal parameters by minimizing the sum of squared errors (SSE) between actual and predicted values.

Question 5 of 8

With $\bar{x} = 6$, $\bar{y} = 65$, $\beta_1 = 6.5$, the intercept β_0 is:

A) 39
B) 26
C) 71.5
D) 6.5

> **Answer: B** — Calculation: $\beta_0 = \bar{y} - \beta_1 \bar{x} = 65 - (6.5 \times 6) = 65 - 39 = 26$.

Question 6 of 8

Using $\hat{y} = 26 + 6.5x$, the prediction for $x = 7$ hours is:

A) 65.0
B) 71.5
C) 78.0
D) 32.5

> **Answer: B** — Calculation: $\hat{y} = 26 + 6.5(7) = 26 + 45.5 = 71.5$.

Question 7 of 8

If $\Sigma(y - \hat{y})^2 = 10$ over $n = 5$ points, the MSE is:

A) 10
B) 5
C) 2
D) 50

> **Answer: C** — Calculation: MSE $= (1 / n) \times \Sigma(y - \hat{y})^2 = 10 / 5 = 2.0$.

Question 8 of 8

An R^2 close to 1 means the model:

A) Explains almost all the variation in y
B) Explains none of the variation
C) Has a huge error
D) Has no intercept

> **Answer: A** — An R^2 value near 1.0 (or 100%) indicates that the regression model accounts for nearly all the variability of the dependent variable.